POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Chemistry

Semester: Spring

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is buffer solution? Write mechanism to show acid buffer solution of benzoic acid and sodium benzoate maintains its P^H constant even after addition of few drop of strong acid or base. Find the P^H of buffer solution which contains 0.1M potassium acetate and 0.2M of acetic acid with2 % degree of dissociation. What will be changed in PH after addition of 2×10^{-3} M KOH solution.(K_a of acetic acid=1.8×10^{-5}) 8

b) How does zinc metal produce single electrode potential? Use SHE to determine single electrode potential copper. Calculate the emf of the cell at 300K from given pairs of half cells. 7

$$Zn(s) \longrightarrow Zn^{+2}(0.1M) + 2e^- \qquad E^0 = 0.76 \text{ volts}$$

$$Ag(s) \longrightarrow Ag^+ (0.01M) + e^- \qquad E^0 = 0.84 \text{ volts}$$

2. a) Give reasons for the followings: 8

i. $TiCl_3$ is coloured but $TiCl_4$ is colourless.

ii. Transition elements show variable oxidation state.

iii. Zn^{+2} salts are white.

iv. Cu is not considered as transition element.

b) Define ionization energy. Explain the factors affecting negativity. How electron affinity of chlorine is higher than Fluorine. 7

3. a) What is Elimination reaction? Differentiate between E_1 and E_2 reaction with suitable reaction mechanism. 8

b) How carbocation are produced? Explain their types along with stability order. 7

4. a) What is polymerization reaction? Write the different types of polymerization with suitable examples for each. 8

b) Explain briefly: 7

a) PVC

b) Vulcanization of rubber

5. a) What is the principle fractional distillation? Write its applications. 5

b) Write properties, preparation and uses of TNT and TNG. 6

c) Mention chemistry and setting mechanism of cements. 5

6. a) What is ozone layer depletion? Mention causes, hazardous affects and control measures of global warming. 8

b) Define alkalinity of water. How is it measure in the laboratory? 7

7. Write short notes on: **(Any two)** 2×5

a) Corrosion

b) Nitration

c) Thermoplastics and Thermosetting plastics